6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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1 Introduction

Clouds are a major component in regulating the Earth's radiative budget and the hydrological cycle. They interact with the solar and thermal radiation affecting the energy balance and the extent of this impact depends on the cloud microphysical and macrophysical properties. However, these properties largely vary depending and therefore the study of clouds becomes quite complex. Clouds present a very large spatial and temporal variability compared to gasses and aerosols (int). Additionally, the phase of hydrometeors within the clouds can vary significantly affecting the cloud radiative properties (Yoshida and Asano).

Partly because of this, clouds remain poorly understood. The lack of data in climatically sensitive regions, such as the Mediterranean basin (Marinou et al.) also contributes to this gap of knowledge about clouds. Several studies have provided statistics on cloud properties covering this region (INCLUIR REFERENCIASXXX). In order to provide information on the vertical structure of clouds, these studies are based on the use of remote sensors such as lidars and cloud radars (Protat et al., a, c). Satellite studies using active remote sensors provide a general picture of cloud properties, but they present low temporal resolution and information on low level clouds is affected by large uncertainties. To overcome these difficulties obtaining more detailed information, ground-based active remote sensors observations are required. here are studies on clouds using ground-based remote sensing measurements in Europe, but they are mostly located in Northern Europe (Stefan Kneifel et al.; Tatiana Nomokonova et al.; Bühl et al.; Peggy Achtert et al.).

However, the information on the Mediterranean region is scarce. Clouds in this region form through various mechanisms, including thermodynamic conditions, Sahara dust intrusions Casquero-Vera et al., and cut-off lows (COLs) systems, which create environments conducive to storms. These complex formation mechanisms pose significant challenges for accurately representing these clouds in atmospheric models. Consequently, there is a critical need for enhanced cloud analysis and data collection in the Mediterranean region. Since 2018, the AGORA (Andalusian Global ObseRvatory of the Atmosphere) ACTRIS CCRES station, in the southeast of the Iberian Peninsula, located in the western Mediterranean basin, has been providing cloud remote sensing measurements. This dataset represents a database for cloud statistics, providing comprehensive data for cloud analysis over an extended period. Notably, it stands as the only cloud remote sensing dataset available in Spain, making it an invaluable resource for regional climate studies. It contributes significantly to filling the gap in cloud data for this climate variability hotspot, enhancing our understanding of cloud dynamics and aiding in the improvement of predictive models and climate studies in these critical areas

This study aims to classify different cloud structures observed over the AGORA station and characterize their physical properties (microphysics and macrophysics). The cloud classification and the statistical quantification of liquid and ice properties (effective radius and mass) are obtained from Cloudnet post-processed data set. The cloud types are divided into liquid, ice and mixed-phase clouds, being precipitative or non-precipitative. Additionally, we focus on assessing a new approach for cloud classification and evaluate its impact on the derived cloud properties. This method classifies clusters of hydrometeors found in Cloudnet masks into clouds. This marks an advancement over previous studies that classify clouds as sequences of hydrometeors by individual profiles. A disadvantage of this approach is the possibility of breaking clouds into different categories, generating high correlation between cloud statistics.

Several studies have already explored the Cloudnet data set for cloud classification by the profile-based method. At the Ny-Ålesund station in the Arctic, this method was employed to characterize single layer clouds and analyze their monthly variability during two years (Tatiana Nomokonova et al.). Similarly, in the German alps, Stefan Kneifel et al. used the same method to classify ice clouds in order to evaluate cloud variability at high altitude locations. Lately, Răzvan Pîrloagă et al. applied a methodology quite similar to that of Tatiana Nomokonova et al. for cloud classification. Then, seasonal cloud variability was achieved using multi-year Cloudnet data in Bucharest, Romania. Overall, these studies used the profile-based method to characterize clouds in the Northerly regions of Europe. Conversely, this study proposes a novel cloud characterization approach in Southern Europe, specifically in the western mediterranean.

The experimental site, datasets, instruments and products are described in Section 2. The proposed new approach on cloud classification algorithm and the cloud assessment are presented in Section 4. Section 5 analyzes the annual, seasonal, and daily cycle of cloud types, as well as the base and top height, and thickness. In addition, this section discusses the liquid and ice properties variability for different clouds. Finally, Section 6 summarizes the main findings and discuss their importance to improve further cloud statistics analysis.

2 Experimental site and Instrumentation

The AGORA ACTRIS-CCRES station (37°2'E, 3°6'S, 680 masl) is located in the Southern of the Iberian Peninsula being one the most meridional CLOUDNET stations. This station is marked by colder months during winter and spring and warm months during summer and fall. Especially between summer and winter, the changes in temperature and humidity are strictly high at this station, as revealed by Andrés Esteban Bedoya-Velásquez et al.. The authors show that these changes are mainly due to solar incidence, being the diurnal cycle a strong component in seasonal differences. For non cloudy data, typically surface temperature in summer can vary around 30 – 40 °C, meanwhile they vary around 15 °C in winter at 16 UTC. For the same time, the relative humidity below 5 km a.g.l presented the opposite behavior, with maximum found between 2 and 10 UTC, and from 9 UTC to midnight. This maximum can range between 60-75% during winter, and 40-50% during summer. However, larger values were also detected above 5 km at 16 UTC within thinner layers for all seasons. The height of this humidity layer can vary with seasons, being higher in summer and fall (7-9 km) and lower in winter and spring (5-7 km). These thermodynamic conditions will be particularly important to understand diurnal cycle of cloud types and inter-annual differences in cloud properties analyzed in further sections.

2.0.1 Instrumentation

In this study we use a combination of active and passive remote sensors that are part of ACTRIS CCRES and exploited in CLOUDNET. These instruments include a MWR, a ceilometer and a cloud radar. An overview of each instrument is provided next. The RPG-HATPRO G2 (Humidity and Temperature PROfiler) is a passive remote sensing instrument that measures the brightness temperature around the water vapor (22-31 GHz) and oxygen (51-58 GHz) absorption bands. This instrument has radiometric accuracy of 0.3-0.4 K and provides accurate values of retrieved liquid water path (LWP) and integrated water vapor (IWV) with 1 s of integration time (Löhnert and Crewell). In addition, its dual-channel system makes it possible to provide temperature and humidity profiles with the same time resolution.

The CHM15k Nimbus ceilometer is an active remote sensing instrument that works with a pulsed laser of Nd:YAG. It measures in the photon-counting mode and provides the attenuated backscattering coefficient (β) of aerosols and small cloud droplets. This instrument has repetition frequency range about 5-7 kHz, where each pulse is emitted at 1064 nm with 8.4 μJ of energy. The laser beam has a range resolution of 15 m and can diverge 0.3 mrad, reaching a complete overlap of about 1500m a.g.l Heese et al.. Especifically, this ceilometer is operated with a time resolution of 15 s.

The dual polarimetric RPG-FMCW 94 GHz Doppler cloud radar is also an active remote sensing instrument that operates at 94 GHz. This instrument uses the frequency-modulated continuous wave (FMCW) signal which makes it possible to vary the vertical and temporal resolution. Their frequency range is from 300-3700 kHz and the antenna beam width is 0.56 °. It measures the Doppler velocity spectrum (DVS) for horizontal and vertical linear polarizations at 94 GHz. Unlike traditional single polarization measurements that retrieve only reflectivity (Z), mean Doppler velocity (ν_D), and spectral width (S_ω), this advanced radar also derives the linear depolarization ratio (LDR), enabling the detection of the melting layer, aerosols, and insects as described by Hogan and O'Connor. Table 1 provides an overview of the AGORA instrumentation, detailing operational frequencies or wavelengths, type of measurements, products, measurement ranges, data collection intervals, and references with detailed descriptions.

Table 1. Summary of instruments and their data products.

Instrument	Bands / wavelengths	Measured	Products	Range	Time	References
94 GHz Cloud Doppler Radar	94 GHz (W-band)	Doppler velocity spectrum (DVS)	ν_D , Z, S_w	[13,51] m	[1.1, 3.6] s	Nils Küchler et al.
MWR RPG HATPRO	22-31 GHz (K-band) 51-58 GHz (V-band)	Brightness Temperature	q and T LWP	[10, 300] m -	[2, 2.5] min 2 s	Francisco Navas-Guzmán et al.
CHM15k Nimbus Ceilometer	1064 nm		β	15 m	15 s	A. Cazorla et al.

This range varies with the height and depends on the chirp table configuration.

It also depends on the chirp type.

Humidity and temperature profiles have a range varying from 10 to 150 m in the first 4.4 km and from 50 to 300 m between 4.4 and 10 km.

Overall, a synergistic combination of these instruments have been widely used for many cloud remote sensing stations as part of the Cloudnet consortium. This combination shows a promising configuration for long-term cloud observations (J. A. et al.), especially to calculate cloud microphysics. Particularly, ground-based radar measurements can be used to retrieve cloud microphysics with better accuracy than satellites (Protat et al., b).

3 Cloudnet database

In order to improve cloud analysis, the AGORA station has been part of Cloudnet since 2018 and was integrated into ACTRIS CCRES (where ACTRIS stands for Aerosols, Clouds, and Trace gasses Research InfraStructure, and CCRES for Centre for Cloud Remote Sensing) in 2023 as a new facility. The Cloudnet processing chain (J. A. et al.) performs extensive quality assurance on raw instruments for providing high-level products, such as target classification, liquid water path, and droplet effective radius. This chain standardizes data processing within the cloud remote sensing community. Their products are derived using a synergistic approach, integrating vertically pointing measurements from ground-based instruments like Doppler cloud radars, microwave radiometers, and ceilometers. Specifically, the target classification product, generated through these three instruments, has been utilized in numerous studies for cloud typing and multi-year analysis (Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.). Therefore, an extensive dataset spanning the years 2018 to 2023 was utilized in this study. Figure 1 illustrates the monthly Cloudnet high level products availability for the AGORA ACTRIS-CCRES station. The availability of this data depends on simultaneous vertically pointing measurements of the microwave radiometer, ceilometer, Doppler cloud radar, as well as the meteorological model products - the ECMWF model (European Centre for Medium-Range Weather Forecasts). As observed, the months with lower data availability are Jan-Mar, with 40-50% of the data available, whereas from Apr to Oct the data availability is over 60% .The years 2018 (270,000 profiles) and 2020 (968,769 profiles) are the years with the least and greatest amount of data. The white bars denote periods with missing data due to instrument maintenance, technical issues, and scanning measurements. Despite the missing periods, the recorded dataset establishes a solid database, covering more than 50

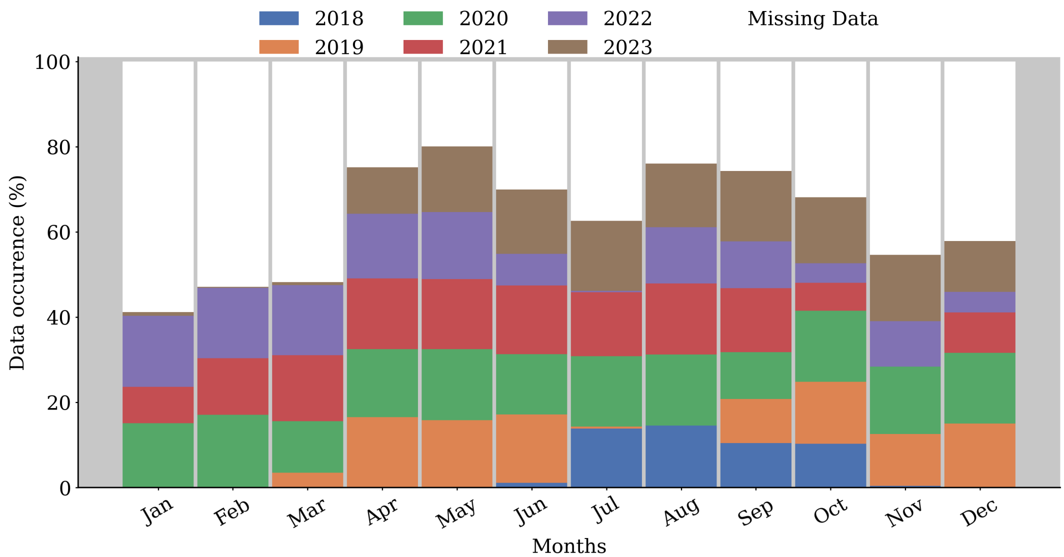


Figure 1. High level post-processed data availability per month at the AGORA ACTRIS CCRES station. This dataset represents more than half a decade of synergic ground-based measurements at this station. Each color represents the data availability for different years.

3.0.1 Data processing

Instrument datasets are processed by Cloudnet processing chain (J. A. et al.) as follows:

1. Set a common grid with a temporal resolution of 30 s and spatial resolution according to the radar (see table 1), respectively. It should be noted that this Cloudnet grid difficulties the monthly profiling on cloud statistics. Hence, another regrinding interpolating onto a height resolution of 20 m is applied in the microphysics analysis.

2. Corrections: this includes two-way gas and liquid reflectivity attenuation, error estimation, and quality flag assignment. High-frequency radars experience significant attenuation from gas and liquid droplets, necessitating signal correction. This issue is particularly pronounced during heavy rain events, where liquid layers on radome sheets can cause an attenuation of 14 dB Hogan et al..

3. Cloud boundaries determination: lidar attenuated backscatter coefficient (β), and radar reflectivity are used. The first is sensitive to small liquid droplets present in cloud bases, as the second are able to penetrate clouds and detect their tops.

4. Target classification product: This product classifies atmospheric particles into different categories by assigning an integer ranging from 0 to 10 at each pixel in the Cloudnet processed grid. Categorizes are as follows: Clear sky (0), Droplets (1), Drizzle or rain (2), Drizzle & droplets (3), Ice (4), Ice & droplets (5), Melting ice (6), Melting & droplets (7), Aerosol

(8), Insects (9), Aerosol & insect (10). These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature (T_w) profiles are used to determine the 0°C isotherm and establish warm and cold regions to identify liquid and ice hydrometeors.

5. Liquid layers respect a threshold of T < -40 °C, since it is not possible to keep water in liquid phase below that temperature. Moreover, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echoes analysis once particle boundaries were already determined. Therefore, falling cold particles were just classified as ice, without distinguishing ice from precipitating ice.

6. Melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 s^{-1} within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on Hogan and O'Connor.

7. The liquid water content (LWC) retrieval assumes a log-normal distribution for warm cloud droplets lifting adiabatically from its base. Its relation depends on atmospheric temperature (T) and pressure (P). This method incurs a relative uncertainty of 15% as shown by A. S. Frisch et al.. At our station, T and P values utilized by the LWC retrieval are provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). The retrieval equation is applied exclusively to liquid pixels within cloud boundaries, which are determined by Doppler radar reflectivity (Z) and ceilometer-lidar attenuated backscatter coefficient. The liquid pixels, as ice, insect and aerosol pixels are given by the cloudnet target classification product (Hogan and O'Connor). Additionally, LWC values are scaled to align with the Liquid Water Path (LWP) derived from the microwave radiometer. The liquid cloud droplet effective radius (r_{liq}) is retrieved by assuming a log-normal size distribution for warm cloud droplets. This method is insensitive to droplet number concentration and the logarithmic spread of the distribution (S. Frisch et al.). Consequently, its primary source of uncertainty is the reflectivity factor error, resulting in a relative error of 15% for r_{liq} retrievals. This relationship is applied exclusively to liquid pixels, where necessary input data, assumptions, relative errors and the studies from which these products are sourced are indicated in Table 2).

8. The ice water content (IWC) retrieval employs an empirical equation as determined by Robin J. Hogan et al., known as the Z-IWC-T relation, which depends on radar reflectivity and temperature. Compared to other microphysical variables, IWC retrievals has the largest relative uncertainties. For the W-band, the relative error is +55% to -35% for temperatures within the range of -20°C to -10°C, and +90% to -47% for T < -40°C. While temperature is a significant parameter in IWC error estimation, Pablo Saavedra Garfias et al. demonstrated that the primary source of uncertainty is the reflectivity factor error. A typical reflectivity error of 3% can result in a 12% error in the ice water path (IWP), which is the integration of IWC over the atmospheric column. Therefore, errors in IWC can be relatively large. The Z-IWC-T relation is applied only to ice pixels within cloud boundaries. Next, the ice cloud particle effective radius (r_{ice}) is obtained following Julien

Delanoë et al., who assumed the same empirical relation for IWC and the visible extinction coefficient (α) derived by Robin J. Hogan et al.. This approach, termed the Z-T relation, depends on both T and Z, and has aThis approach, termed the Z-T relation, depends on both T and Z, and has a relative uncertainty of 50% of r_{ice} (Hannes Griesche et al.). In general, these high relative errors emphasize how challenging it is to retrieve ice properties accurately (see Table 2)

Table 2. Cloudnet microphysics products used in this study, their inputs and assumptions

Cloudnet products	**Input role**	σ_{re}/r_e	**References**
Liquid water content (LWC)	Z and β: cloud base and top identification T, P, assuming an adiabatic lifting MWR or CDR LWP to scale LWC	15%	Frisch et al. (1998)
Ice water content (IWC)	Z and T profiles	-35% to 90%	Robin J. Hogan et al. Julien Delanoë et al.
Droplet effective radius (r_{liq})	Z, and N and σ of lognormal PSD	15%	S. Frisch et al.
Ice effective radius (r_{ice})	Z and T	50%	Hannes Griesche et al.

4 Methodology

4.1 Cluster classification method

The cluster-based algorithm is developed in order to homogenize cloud clusters in time. Different from the profile-based method, this method avoids high correlation between distinct cloud categories (e.g. ice and mixed-phase). In both cases, the Cloudnet target classification product is used as a basis for cloud typing. This product provides the hydrometeor type information for each profile. Thus, the pixels classified as hydrometeors, such as Droplets, Ice and Drizzle & Droplets are used to classify clouds. In the case of the profile-based method, Bühl et al. already adapted it for only mixed phase clouds, selecting only cases with more than 15 min of coherent cloud profiles (with similar cloud profiling within 300 s). Nevertheless, by adding new criterias to identify liquid and ice clouds, this method can classify segments of one cloud in different categories, depending on the time threshold. Hence, we introduce a new approach that does not classify cloud profiles neither segments, but cloud clusters.

Prior to the use of the target classification product, a preprocessing of the data was required. It has been observed that for some specific cases, misclassification of some pixels may occur. Figure 2 illustrates a thinner and lower precipitable ice cloud present between 08:00 and 20:00. However, there is no evidence of clouds in the attenuated backscatter and the radar reflectivity. These misclassifications stem from an inaccurate representation of the temperature profile by the models in the region of Granada, affecting the target classification algorithm, and subsequently, the cloud classification performed in this study. To mitigate these issues and prevent misinterpretation of future cloud statistics, we established a filter to detect and remove these cases. Basically, all these clouds which were identified with mean cloud base below 4 km (above mean sea level) and average cloud thickness below 700 meters were reclassified as noise. These values were determined empirically by studying many cases separately.

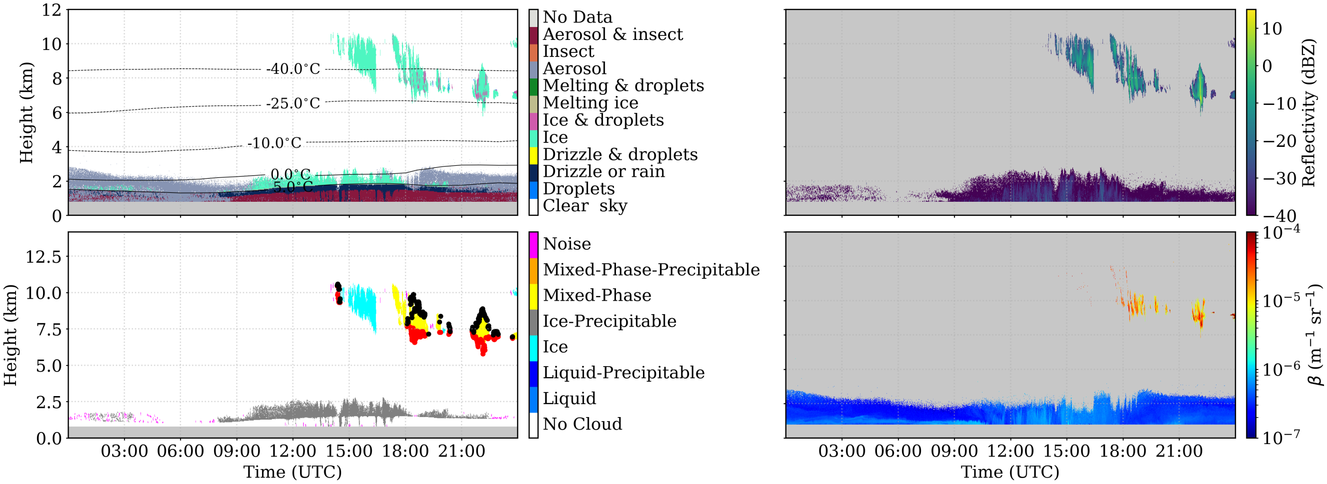


Figure 2. A case study of Cloudnet hydrometeor misclassification. (a) Target classification product; (b) Radar reflectivity factor from Doppler cloud radar; (c) Attenuated backscattering coefficient from ceilometer. No evidence of clouds in the ABL can be identified by the instruments.

Once the data are curated, the different cloud types are identified. Recent studies employed a cloud classification algorithm by profile with the aim of classify clouds (Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.) and analyze their statistics. However, this algorithm admits different cloud typing inside the same cloud as a whole, assigning similar physical properties for different clouds. In the appendix A, we describe this algorithm, adapting the criteria used by Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.. In this study we propose a new methodology to overcome these inaccuracies. We define a cloud as a group of at least 100 pixels identified by Cloudnet as droplet, drizzle (see Figure 3). To do so, a cloud mask is generated from the CLOUDNET target classification. Cloud pixels separated only by two pixels in any direction (analogous to 1 min, and 20 - 50 m of height) are considered part of the same cloud. This is achieved using a pixel dilation and contraction technique (Dougherty), which expands and contracts the cloud shapes (see Appendix B for an explanation of the method). Once the cloud pixels are grouped, each cloud must have more than 100 pixels to be considered (without considering rain such as Drizzle or Rain pixels). Otherwise, it is deemed as an unknown type (likely noise). In order to illustrate this method, the left panel of Figure 3 shows cloud clusters of different colors during an entire day (on 11 February 2021). The regions identified as clouds are marked by the black dashed lines in the right panel of the same figure, where clusters that are very close will merge to form one cloud. Also, small and isolated groups with less than 100 pixels are indicated by the red dashed lines

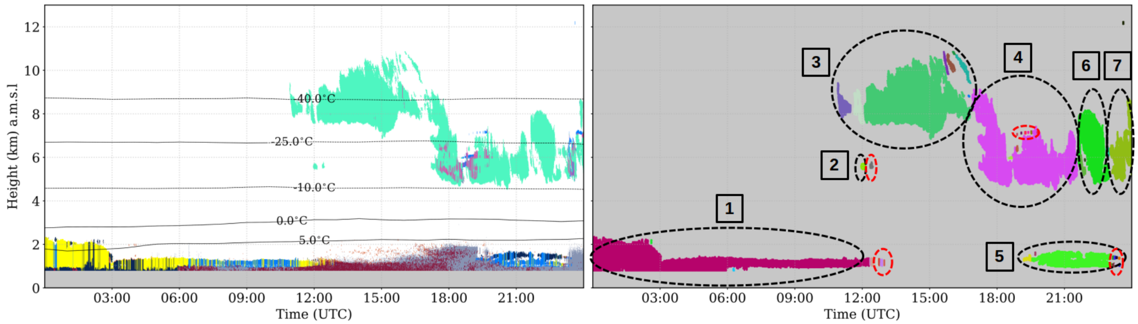


Figure 3. Cloud cluster identification by Cloudnet target classification on 11 February 2021. (a) All hydrometeor clusters identified at different colors; (b) Cloud clusters identified in black dashed lines and noise by red dashed red

Then, clouds are classified in six types: liquid, ice, mixed-phase, precipitating liquid, precipitating ice, and precipitating mixed-phase. To do so, the hydrometeor type percentages are calculated for each cloud, (shown in figure 4). A cloud is classified as liquid cloud (Liquid criteria) it comprises more than 70% of Droplets and Drizzle plus Droplets. Then, it is further classified as a precipitating liquid cloud (Rainy criteria) if it contains at least 10 connected rain pixels, Similarly, a cloud is classified as an ice cloud (Ice criteria) if the previous verification is not satisfied and if this group contains more than 90% of ice or less than 10% of Droplets plus Ice & droplets. The same rain verification as in the liquid case is made to classify this group as a pure ice cloud or a precipitating ice cloud. Otherwise, clouds are classified as mixed-phase or precipitating mixed-phase through the same rain verification as in the other clouds. In relation to precipitating ice and precipitating mixed-phase clouds, an extensive analysis of individual case studies revealed that Drizzle & droplets only appear below the melting layer together with Drizzle & rain droplets. Therefore, to avoid an underestimation of cloud base height due to drizzle, we removed these pixels for these clouds since they are probably rain.

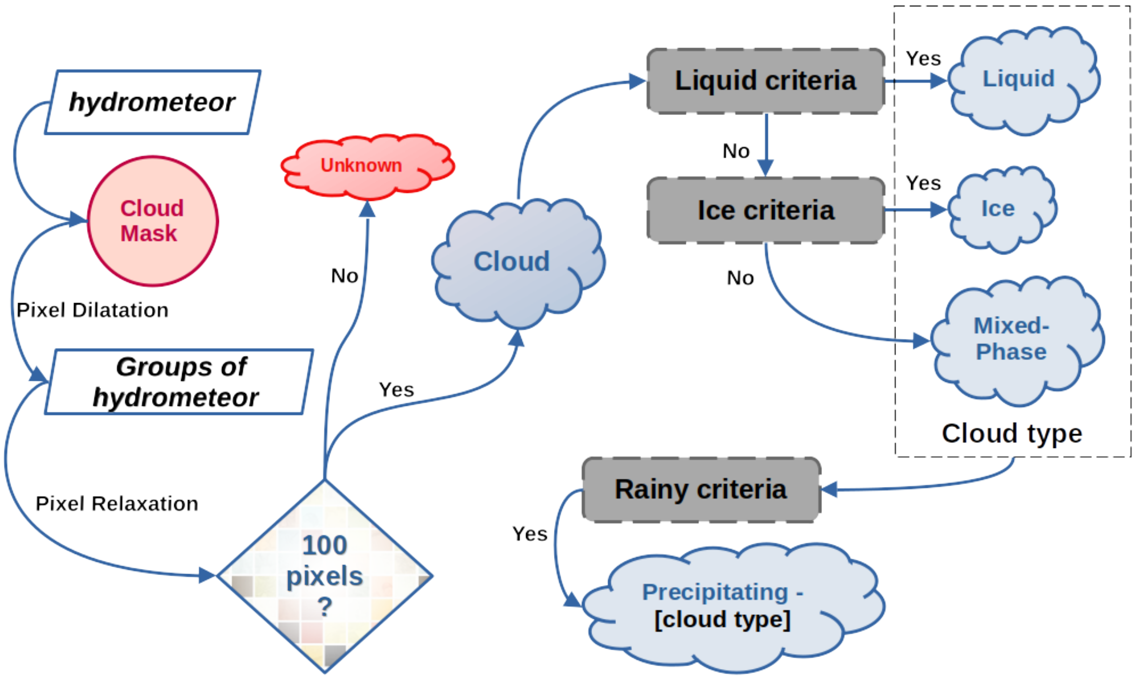


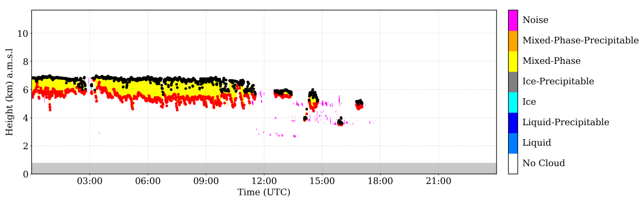
Figure 4. Scheme of the cloud classification algorithm by hydrometeor grouping

After the cloud classification, each cloud is then classified as single or multi-layer over time. Initially, time intervals where clouds overlap are identified. Overleaped pieces of clouds are classified as multi-layer, whereas non-overlapping pieces in principle are classified as single layer.We then analyzed cloud profiles for each cloud individually, acknowledging that a cloud can exhibit multiple layers with itself. If a cloud had a second layer separated by only one pixel at some time interval, that interval is identified as containing multi-layer. Otherwise, the clouds at that interval are classified as a single layer.

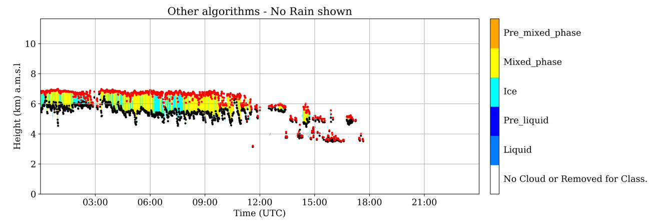
In this study, only single layer clouds are analyzed. For the single layer case, cloud properties are calculated as follows: the cloud base and top height are determined by the first and last height pixels within the cloud layer, respectively. Cloud thickness is obtained by the difference between cloud base and top height pixels.

4.2 Cloud typing assessment

An evaluation of the classification algorithm presented in this study against the method used in the literature is conducted in order to evaluate its applicability for further analysis. Here, we aim to provide insights into the correlation on cloud statistics between different cloud categories generated by these two methods. In this framework, Figure 5 shows a case study on 6 May of 2023 in Granada comparing profile-based cloud classification and cluster classification for a mixed phase cloud, where red and black dots represent the cloud base and top, respectively. As expected, the profile-based classification on the right panel leads to the fragmentation of a single cloud into different types. In this case, the same cloud is being classified as ice and mixed-phase by time. On the other hand, in the left panel, the new classification method preserves the homogeneity of the cloud and classifies it as mixed-phase cloud, resulting in a more realistic classification.



(a) Cloud classification using the new method.



(b) Cloud classification using the old method.

Figure 5. Temporary figure (needs target classification). Comparison of cloud classification methods. The left panel illustrates the new method, which preserves cloud homogeneity and provides a more realistic classification. The right panel shows the old method used in recent studies, where cloud classification is time-based.

As a consequence, the profile-based method of cloud classification can generate high correlations between cloud categories in terms of frequency of occurrence and cloud properties. Regarding cloud occurrence, the right panel of Figure 5 suggests that both ice and mixed phase clouds will have similar daily cloud occurrence and average cloud base. However, this is occurring because of the classification method, which could further affect cloud statistics.

Therefore, in order to quantify the correlation between cloud categories generated by these two classification algorithms, we calculate the daily cloud occurrence to assess the correlation between cloud types. Figure 6 shows the pearson correlation coefficients of daily cloud occurrence between cloud types for the cluster classification on the left, and for the classification by profile on the right. As it can be seen, the new proposed method demonstrates generally weaker correlations among cloud types, with most values clustering around zero. For example, the highest correlations observed were 0.19 between Liquid and Liquid-Precipitable clouds and 0.17 between Liquid and Mixed-Phase-Precipitable clouds, indicating relatively weak relationships. Additionally, some negligible negative correlations were observed, e.g. between Ice and Mixed-Phase-Precipitable clouds (-0.1). This suggests that the new method may provide a clearer distinction between cloud types, which is advantageous for analyses requiring precise classification. In contrast, the profile-based classification exhibited stronger and more pronounced correlations. Notably, the correlation between Ice and Mixed-Phase clouds was 0.4, and between Liquid and precipitating liquid clouds was 0.32, highlighting significant interdependencies between these cloud types. These relatively higher correlations are a consequence of the classification scheme defined in this method, which allows tagging multiple cloud types inside the same cloud domain. As a result, the comparison indicates that the new method, with its generally weaker correlations, may be better suited for distinguishing different cloud types. These findings underscore the importance of selecting an appropriate analytical method to study cloud properties between different cloud types.

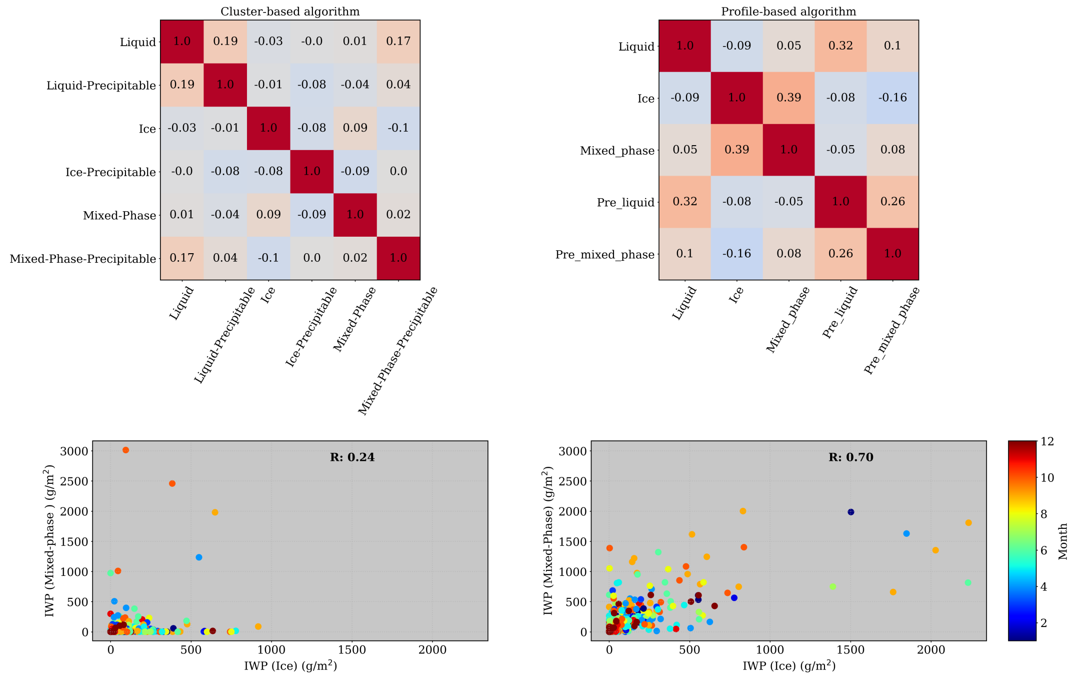


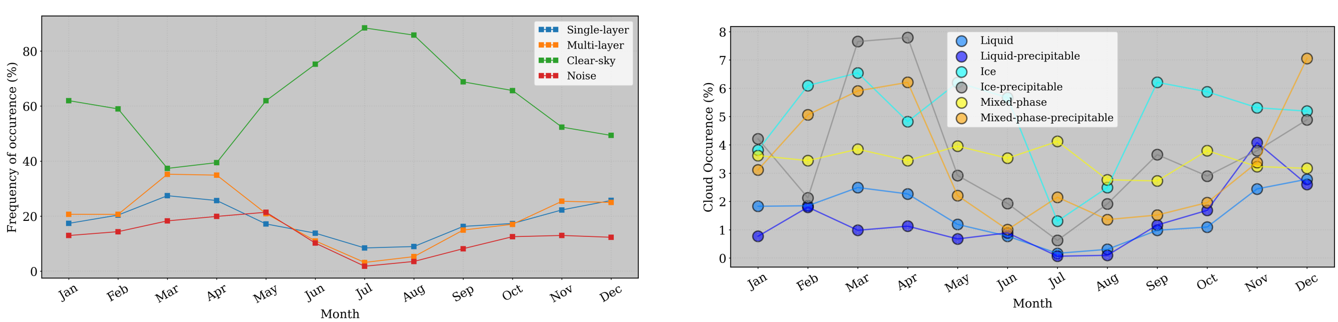
Figure 6. The Pearson correlation matrix of daily cloud occurrence among different cloud types is shown in the top panel. The correlations for cluster-based methods are on the left and the profile-based correlation is on the right. Below each correlation matrix, the corresponding comparison of IWC between ice and mixed phase is shown.

Regarding the correlation in cloud properties, we computed the daily average of IWP for ice and mixed-phase clouds for the two classification methods. The bottom panel of Figure 6 shows the IWP of mixed phase clouds in function of the IWP of ice clouds. The R factor is the Pearson correlation coefficient between the IWP of these two cloud types. This coefficient is quite higher for the profile-based method, showing a correlation of 0.7. On the other hand, the cluster method exhibited only 0.24, with no apparent correlation. In addition, the correlation seems to be not monthly dependent as there is no clear separation by month. It is worthy to note that this section is not focused on the analysis of IWP, this will be done in section 5.3. Instead of this, here we focused on the influence of the classification algorithm on cloud properties analysis. Following, the observations suggest that statistics on cloud properties can be highly influenced by the profile-based classification. As a result, this method may lead to the misinterpretation of real cloud properties, raising the relevance of previous evaluation of cloud classification algorithms before doing statistics on cloud properties. In this light, the new approach shows a promising classification method and will be employed in this study.

5 Results and discussion

5.1 Annual variability and daily cycle of clouds

This section aims to analyze daily and seasonal variability of cloud frequency of occurrence at the AGORA station. Figure 7a presents a first overview on cloud occurrence, showing the monthly occurrence frequency for different sky conditions, i.e. single-layer clouds, multi-layer clouds, clear sky, and noise (as defined in section 4.1). The data reveal a distinct seasonal pattern in cloud cover and clear sky occurrences. Clear sky conditions (green line) are predominant all year long, being most frequent in the summer months, peaking in July and August (> 80%), with a significant decline in spring, reaching the lowest point in March and April (40%). This pattern indicates a higher prevalence of clear skies during the warmer months and increased cloudiness during the colder months, according to section 2 Single-layer clouds (blue line) exhibit a relatively weak seasonal pattern throughout the year, with a maximum in spring (35%) and a decrease in the summer months (10%), suggesting a considerable variation in single-layer cloud occurrences. Multi-layer clouds (orange line) show a similar pattern, with a maximum in March and April and a reduction during the summer, indicating a predominance of multi-layer clouds in the spring, with more occurrence of multi-layer than single-layer clouds. The noise category (red line) maintains a weak seasonal pattern throughout the year, comparable with multilayer in later spring and summer. Overall, the data highlight clear seasonal variations in sky conditions, with predominating clear skies throughout the year and increased cloudiness, both single and multi-layer, in spring and winter. Although AGORA exhibits a considerable multi-layer occurrence in spring, these clouds are problematic due to inaccurate LWP partitioning among cloud layers (Shupe et al.). The partitioning needs an accurate cloud profile, which cannot be assessed for elevated clouds with only radar reflectivity. The reflectivity is not sensible to small cloud droplets and the lidar signal is totally attenuated by the first layer. Hence, liquid layers will be poorly identified causing an inaccurate determination of liquid and mixed phase clouds as observed by Tatiana Nomokonova et al.. Therefore, the following analyses are focused only on single-layer clouds, seeing their physical properties can be retrieved more accurately.



(a) (b)

Figure 7. On the left, inter-annual frequency of occurrence for single layer clouds (blue), multi-layer clouds (orange), clear sky (green) and noise (red). Right panel, inter-annual single layer cloud occurrence for different cloud categories (see section 4.1). Each cloud types is indicated in the legend

Monthly occurrence of single layer clouds is depicted in Figure 7b. The percentage of occurrence in this figure is calculated in relation to the total number of data. Thus, the sum of cloud occurrences per month must retrieve the orange curve in the left panel of Figure 7a. This percentage is defined in relation to the total number of single layers. Mixed-phase clouds (yellow) exhibit a relatively consistent occurrence with slight fluctuations around 3% and 4%. Liquid-precipitable (dark blue) clouds have a modest variation with zero occurrence in summer, followed by an increase during the transition to winter, reaching a maximum of 4% in November, suggesting increased precipitation during fall. Liquid clouds (blue) have quite a similar percentage, with 3% in March and December, against zero in summer. Ice clouds (cyan) drop sharply in July and August and maintain almost constant levels around 5% and 7% throughout the rest of the year, indicating that hotter temperatures inhibit ice cloud formation during summer. Ice-precipitable clouds (gray) have a notable peak of 8% in spring, followed by a decrease in summer (1%) and an increase in fall, until they reach a local maximum of 5% during winter. It suggests seasonal variations in ice-related precipitation with maximum in spring, unlike the case of water-related precipitation, which showed a maximum in November. Mixed-phase-precipitable clouds (orange) peak in April (6%) and December (7%), indicating higher chances of mixed-phase precipitation towards spring and the end of the year. Overall, ice, precipitating ice, and precipitating mixed-phase clouds have the strongest seasonal pattern, with a predominance of these types of clouds over liquid clouds throughout the year. Moreover, the precipitation in this region is intrinsically related to ice and mixed phase clouds, occurring mostly in spring During summer, non-precipitating clouds are observed. Together with the elevated summer temperatures, the Iberian Peninsula experiences a large low-pressure area over northern Africa, with high-pressure systems over the Atlantic Ocean and eastern and central Europe that results in easterly winds over Andalusia, transporting moisture from the Mediterranean Sea in summer (F. J. Bello-Millán et al.). This combination of high temperatures and humidity can create an environment conducive to storm formation in Granada during the summer. In this context, the few clouds observed in summer are likely related to convectively formed cumulus or cumulonimbus clouds.

The comparison of cloud daily cycle clouds per season was also performed. Howere, since no marked diurnal cycles were observed at any season, this comparison is not shown. Overall, liquid and liquid-precipitable clouds are the least likely, independently of the daily hour. Differently, slight variations of mixed and ice clouds in spring and fall were observed, where during fall, ice clouds fluctuated around 6-8% between 08:00-14:00 UTC. On the other hand, other single layer clouds have less than 4% of occurrence at this time. In spring, ice, mixed-phase and ice-precipitable clouds present a higher occurrence. Spring shows a slight peak of occurrence for Ice-precipitable and Mixed-phase-precipitable clouds between 09:00 and 21:00 UTC. Different from non-precipitating clouds which reached a minimum around mid-day, these clouds doubled the occurrence from at 00:00 (4%) to 16:00-17:00 UTC (8%). Despite that these observations presented small diurnal variations, they are consistent with the findings of Shahi et al., where convection-permitting models of the Iberian Peninsula (IP) captured high values of mean precipitation during spring. Additionally, Alexandre Rios-Entenza et al. revealed that specific synoptic conditions, such as a reduction in the strength of the zonal circulation, can favor local and mesoscale convection during spring in the western region of the IP. This analysis confirms that the maximum cloud and rain occurrence observed during peak sunlight hours may be strongly associated with local convection.

5.2 Seasonal patterns in cloud base, top, and thickness

Seasonal evolution of cloud base height, top height and thickness are displayed in Figure 8. Each color represents a cloud type, where liquid clouds are in blue, ice clouds are in gray and mixed-phase clouds are in orange. Also, the lightest colors represent non precipitating clouds, and the darkest colors represent precipitation clouds. The cloud base analysis reveals that precipitating clouds exhibit a significantly lower cloud base height (top panel) compared to non-precipitating clouds, as well as a narrower distribution. Regarding non-precipitating ice clouds, they possess the highest cloud base heights, with median values around 8 km. The ice clouds with base height above the 50/75th percentile can be normally associated with cirrus clouds. Next, non-precipitating mixed-phase clouds have the second highest cloud base, showing considerable variability, except during summer, when values stabilize at 5.5 km. Also, these clouds present a higher base in fall, where the 50th percentile reaches 6.5 km. Differently, the precipitating mixed phase clouds have much less variability and higher cloud base at 4.5 km in summer, as expected. For non precipitating liquid clouds, cloud bases are highly variable during summer, achieving significantly different heights compared with other seasons. Overall, in summer, precipitating mixed-phase clouds, and ice precipitating clouds demonstrate elevated cloud base heights, attributed to the relatively high temperatures and low humidity at this season. Liquid clouds also have elevated base height in summer, however only a few cases were registered, with 50th and 75th percentiles around 4 km and 6 km, respectively. These clouds can be related to stratocumulus and stratus clouds, which were decoupled from cumulonimbus at the final stage in summer.

In general, precipitating clouds also have significantly lower median cloud top height (panels at the center of Figure 8) compared to non-precipitating clouds and exhibit a narrower cloud top distribution, except in the case of ice clouds. Non-precipitating ice clouds have median cloud tops height around 9.5 km, while ice-precipitable clouds demonstrate highly variable cloud tops. Given their well-defined cloud bases, these clouds exhibit greater cloud thickness, indicative of storm activity at our station. The cloud tops for liquid clouds mirror the variability of their cloud bases, especially during summer. These clouds exhibit a well-defined cloud thickness, with non precipitating liquid clouds having a thickness around 150 m and precipitating liquid clouds reaching up to 500 m. Non precipitating mixed-phase cloud tops also present high variability. Nevertheless, they displayed a well-defined cloud thickness, with similar distribution along seasons. Their median cloud thickness is around 400 m, increasing to 500 m for the precipitating case (dark orange), with slightly higher values in summer. In general, non-precipitating mixed-phase clouds have median cloud tops around 7 km during spring, summer, and winter, but reach 8.5 km in fall. Then, precipitating mixed-phase clouds have cloud tops around 3 km in spring, fall, and winter, peaking at 7 km in summer.

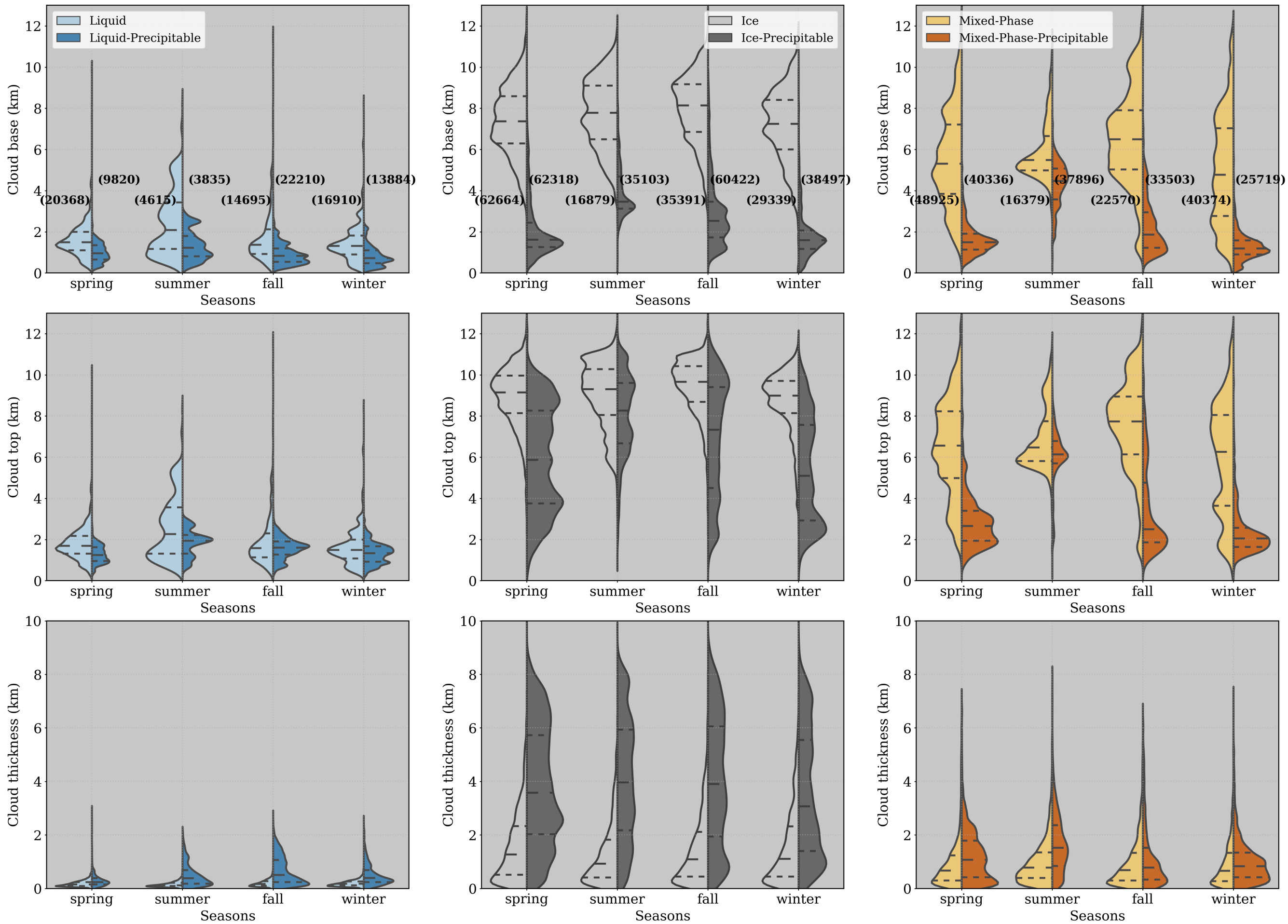


Figure 8. Seasonal evaluation of cloud base height (top), top height (middle) and thickness (bottom). For different cloud types: liquid (left), ice (middle) and mixed-phase (right)Dashed lines indicates 25/50/75th percentiles

Table 3 resumes the median cloud base and top heights found in the figure 8. Overall, this table shows that ice-precipitable clouds exhibit the largest cloud thickness, followed by Ice, mixed-phase-precipitable, Mixed-phase, Liquid-precipitable, and Liquid clouds. Ice clouds present the highest cloud base height, followed by Mixed-phase, Ice-precipitable, Mixed-phase-precipitable, Liquid and liquid-precipitable. It is important to note that, despite Cloudnet post-processed data being corrected for liquid attenuation, cloud tops for precipitating clouds can be underestimated due to high liquid attenuation.

5.3 Cloud microphysical properties

In this section we investigate the inter-annual variability of microphysical retrievals for each category of single-layer clouds. Clouds were divided into Liquid, Mixed-phase and Ice categories, where inside each category non precipitating and precipitating cloud properties are discussed. The assessed cloud properties are liquid water content and droplet effective radius for liquid

Table 3. Cloud macrophysics per season and cloud type. Median values of cloud base height, cloud top height and cloud thickness are shown

		Seasons			
Cloud Properties	**Cloud Type**	**Fall**	**Spring**	**Summer**	**Winter**
Cloud base height (km)	Liquid	1.37	1.49	2.09	1.31
	Liquid-Precipitable	0.83	0.95	1.22	0.72
	Ice	8.14	7.36	7.78	7.24
	Ice-Precipitable	2.53	1.61	3.46	1.60
	Mixed-Phase	6.50	5.31	5.49	4.77
	Mixed-Phase-Precipitable	1.86	1.49	4.47	1.19
Cloud top height (km)	Liquid	1.58	1.70	2.27	1.49
	Liquid-Precipitable	1.61	1.25	1.94	1.34
	Ice	9.66	9.15	9.30	8.99
	Ice-Precipitable	7.33	5.87	8.26	5.10
	Mixed-Phase	7.73	6.56	6.47	6.26
	Mixed-Phase-Precipitable	2.50	2.65	6.14	2.06

and mixed phase clouds, ice water content and ice effective radius for ice and mixed phase clouds (of precipitating and non precipitating type). Since cloudy conditions are not common at the Granada station, the median profiles were used in order to avoid the effect of outliers in the annual variability of cloud properties.

5.3.1 Liquid clouds

The monthly median profile of LWC shows high variability of LWC in function of height for liquid clouds (Figure 9. LWC is generally around 50 mg m^{-3} during the whole year, with no seasonal pattern. Maximum median values are observed during summer, fall and winter, whereas for spring a maximum is observed between 4 and 5 km. A sharp peak of about 500 mg m^{-3} can be also clearly identified in spring below 500 m. This peak is related to liquid clouds observed during the springs of 2020, 2021 and 2022. For particular months such as February, March and April, quite large values below 1 km and within 2.5 and 4 km (right panel of Figure 9) are observed. Nevertheless, above 500 m, really high values observed in the monthly profiles are not representative of the entire season as can be seen in the left panel of Figure 9. Summer months exhibited a lack of liquid phase clouds, with some elevated LWC at higher levels as shown in the left panel of Figure 9. The average liquid water path (LWP) (bottom panel of Figure 9, black star) shows a considerable variability of LWP. It highlights a very wide variety of liquid clouds in these months. A notable increase in the median LWP occurs in April, aligning with their maximum cloud occurrence. From August to October, there is a gradual increase in average LWP, while the median stays stable. This trend corresponds with the increased occurrence of liquid clouds from summer to fall, impacted by different mechanisms of cloud formation. Since these months reported just a few occurrences of liquid clouds, as can be checked by the blue line, which shows the number of cloud profiles per month, their statistics will be much more sensible to these mechanisms. As a result LWP in summer is the smallest, and there is no significant difference between the average and the median.

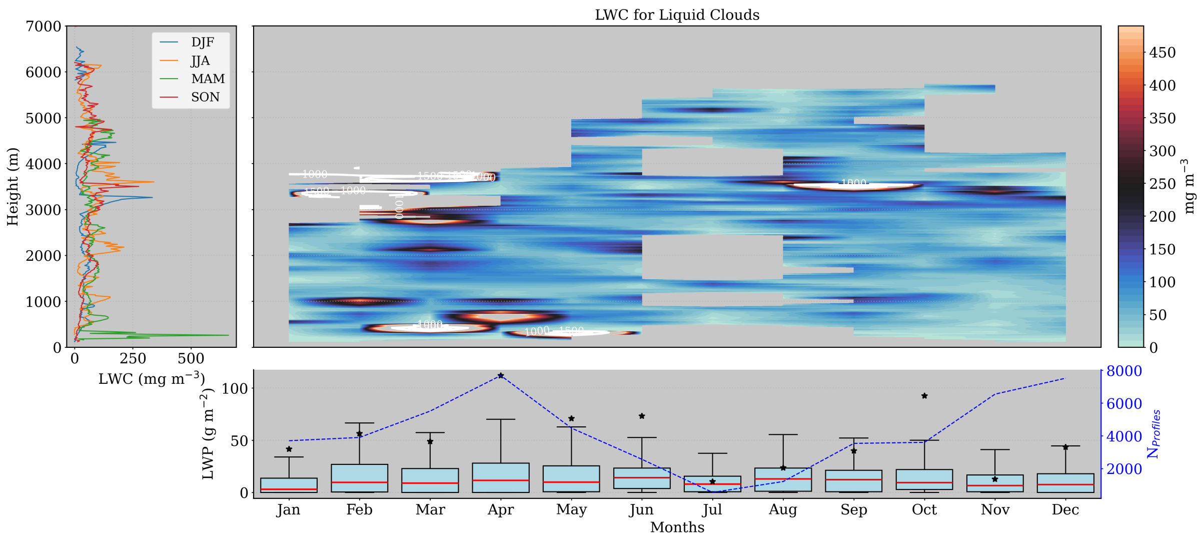


Figure 9. Seasonal and monthly average of liquid water content (LWC) and liquid water path (LWP) for liquid clouds. Left panel shows the seasonal median profiles of LWC. Right panel shows the monthly median profiles of LWC and monthly box-plot of LWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

In terms of droplet effective radius, liquid clouds are well consistent throughout the year, showing small differences in terms of median values (top panel of Figure 10), mostly ranging between 5 and 8 μm. There is no discernible pattern in r_{liq} across seasons, with slightly larger droplet sizes at higher altitudes. In addition, there are almost no liquid droplets above 4 km, with just a few occurrences in summer and early fall registered. The median r_{liq} is generally stable around 5 μm, with a small decrease in July, when almost no clouds were registered. The seasonal profiles confirm the uniformity in r_{liq} distribution with height, showing minimal variation with altitude. It suggests a consistent cloud composition and droplet size throughout different atmospheric conditions. These findings underscore the stable nature of liquid clouds in terms of droplet size and LWC. Normally, these clouds have a small thickness, as observed in Fig XXX, associated with a small range of r_{liq}

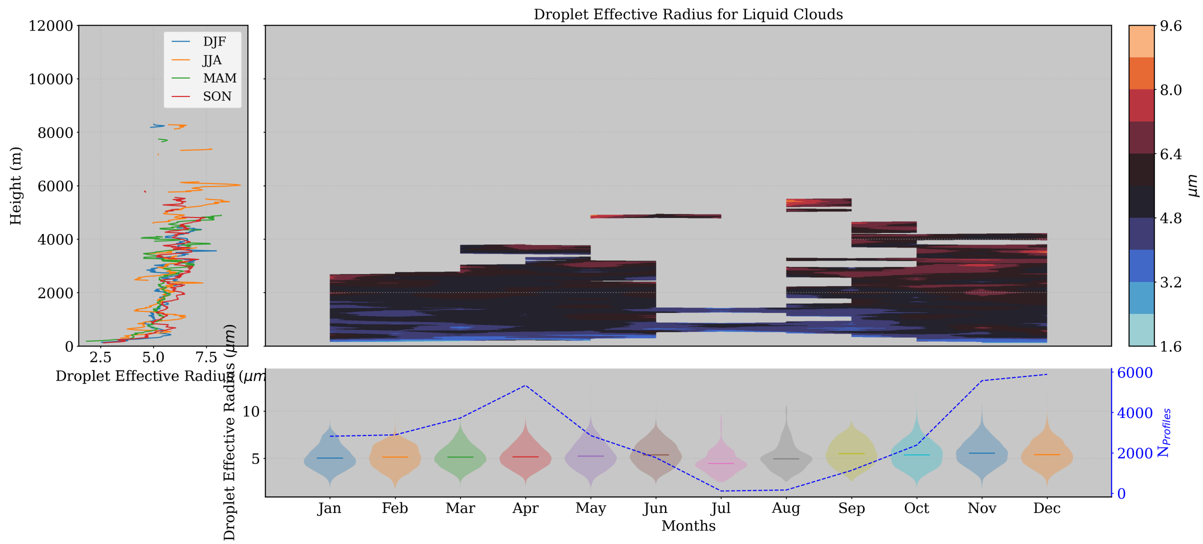


Figure 10. Seasonal and monthly statistics on droplet effective radius (r_{liq}) distribution for liquid clouds. Left panel shows the seasonal median profiles of droplet effective radius. Top-right panel shows the monthly profiles of the median r_{liq}, while the bottom-panel displays the size distribution of r_{liq} by months.

Precipitating liquid clouds exhibit distinct seasonal vertical profiles of LWC. The left panel of Figure 11 shows that LWC can reach more elevated heights during spring and fall. In general, this parameter increases with altitude up to 2 km, with the exception of spring, which maintains relatively constant values within this layer. At the top of this layer, LWC is typically 250 mg m^{-2}, and approximately 500 meters higher (2.5 km) during the fall. Above this level, LWC is highly variable with altitude in spring, summer, and fall. In particular, summer demonstrates a second maxima of approximately 250 mg m^{-3} close to 3 km, and a sharp peak near the surface. This is also observed during spring due to the presence of drizzle particles below cloud base during rainfall. It has been attributed to relatively humid months in April of 2020, 2021, and 2022 as already observed for non precipitating liquid clouds. The monthly evaluation on the right-top panel of Figure 11 supports this idea, in addition to attributing June ($\sim$300 m) as the month causing the sharp peak in summer close to the surface. For the rest of summer and early fall, atmospheric water content is almost zero. For fall, the peaks observed in 2.5 km occur in October, whereas winter peaks at higher levels occur in Nov-Dec and at lower levels in January. Then, in the rest of summer and early fall, atmospheric water content is almost zero. In terms of LWP, the bottom panel of Figure 11 indicates substantial variability of LWP in January, suggesting a diverse variety of liquid clouds during this month. In contrast, spring exhibits relatively consistent LWP across its months, reflecting stable months for precipitating. In these months, their median stays below 50 g m^{-2}, and summer has even less LWP, except for June, as previously observed.

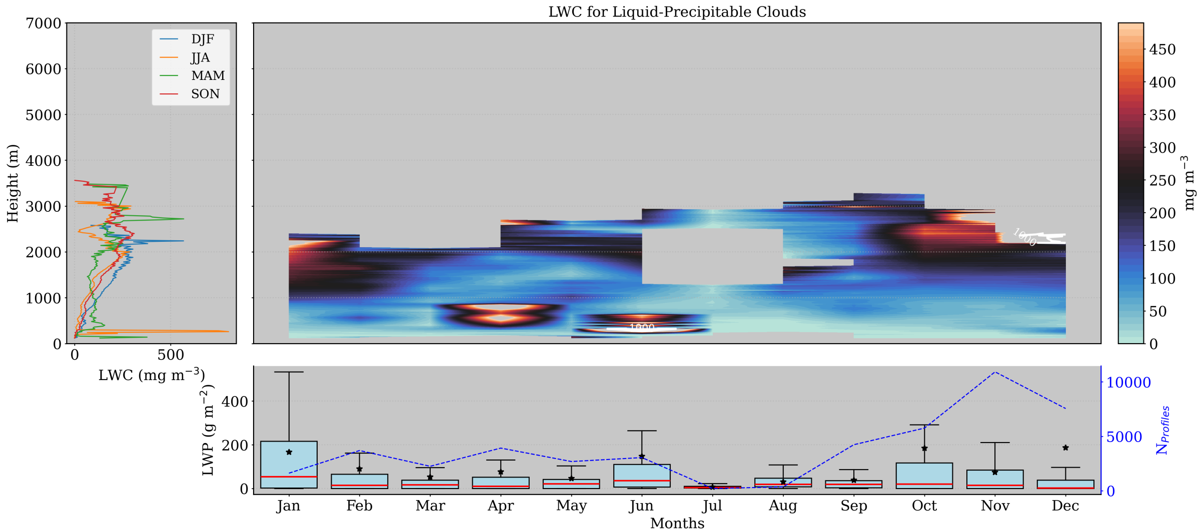


Figure 11. Same as Figure 9, but for precipitating liquid clouds.

Different for the purely liquid clouds, precipitating liquid clouds demonstrated radically different seasonal profiles of median r_{liq} (see left panel of Figure 12). Spring and summer have similar r_{liq} up to 1 km of height, fluctuating around 5 μm. Above this level, summers are still increasing up to 2 km, while spring keeps r_{liq} in 5 μm until 3km, followed by sharp increase to 20 μm. Winter has the second largest r_{liq} at the first 2 km ($\sim$8 μm), disregarding large drizzle particles close to the ground. These particles cause sharp peaks near the surface in summer, fall and winter. In addition, r_{liq} in fall are the largest (below 3 km), reaching 15 μm around 1 km. The monthly profiles in the right panel of Figure 12 shows that larger values in this season occur in November and close to the ground. These sizes are normally related to drizzle droplets as already observed. Their monthly size distribution in the bottom panel of Figure 12 displays different size distributions over the year, where higher median values are found in fall as already observed, specially in November. During spring and early summer, the median is maintained around 10 μm, then decreases in summer, and increases again in fall as expected.

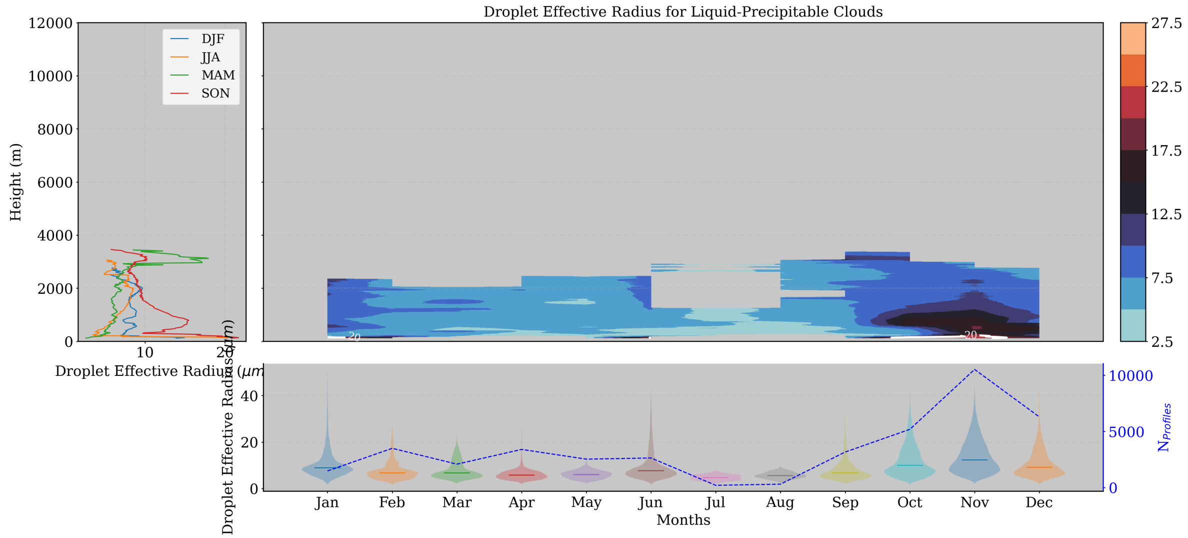


Figure 12. Same as figure 10, but for precipitating liquid clouds.

Overall, Liquid clouds are challenging to analyze due to their low frequency of occurrence. They exhibit significant variability with height, making it difficult to discern seasonal differences in liquid water content (LWC) profiles, except at low levels. Near the surface, LWC is much higher in spring, and their droplet size (r_{liq}) remains relatively constant around 5 μm. In contrast, precipitating liquid clouds display a more pronounced seasonal pattern, with higher LWC and larger r_{liq}. Notably, the presence of drizzle particles near the surface makes r_{liq} larger. During summer, these clouds exhibit two distinct peaks in LWC in function of height, one at 2 km and another at 3 km

5.3.2 Mixed Phase clouds

The LWC for Mixed-phase clouds vary greatly with height for all seasons (see left panel of Figure 13.a), except between 4 and 10 km, where its value is almost zero. However, this variability is masked due to the presence of sharp peaks during spring and summer close to the surface. These peaks are located around 500 m, and reach values of 500 and 1000 mg m^{-3} (even greater than the maximums found for liquid clouds). Next, the right panel of this figure also shows a great monthly variability, with extremely high values of LWC at different levels. Despite the fact that these values are more frequent for mixed-phase clouds than liquid clouds, the integrated LWC (bottom panel of Figure 13a) of both is quite comparable, being slightly larger for liquid clouds. It shows that mixed-phase clouds have even larger variability in LWC, while keep their frequency of occurrence almost constant (around 3-4%) throughout the year. This variability can also be checked by the monthly difference between the mean (black star) and the median (red line) LWP in the bottom panel.

In terms of droplet effective radius r_{liq} (Figure 13.b), the opposite behavior is observed in function of height. The left panel of Figure 13.b show droplet size increasing up to 5 km, then starting to decrease. This profiles highlights a general behavior that can be explained as follows: As temperature decreases with height, ice particles start to form, increasing competition for water vapor and halting the growth of liquid droplets to form ice particles as temperature continues to decrease. It will decrease the size of water droplets keeping their LWC fixed (close to zero between 4 and 10 km) while the ice mass concentration increases (see left panel of figure 13.c). The right panel of Figure13.b allows a more detailed evaluation of r_{liq} per month. Large droplet sizes are located between 4 and 8 km, except for July, which also shows larger droplets at 10 km. As discussed, liquid water is uncommon at these levels, suggesting the presence of clouds with large vertical development as cumulus clouds. Overall, summer exhibits larger droplets compared to colder months, with an effective radius between 14-22 μm at medium heights (> 4 km and < 6 km), and between 12-18 μm around 10 km. Conversely, droplet sizes are generally close to 10 μm, as is shown by the monthly distribution in bottom panel of Figure13.b, where the modes are at even smaller values than the median. In fact, cloud droplets are slightly larger in summer, with a good probability of reaching 15 μm, but generally, they present smaller droplets.

Regarding the ice water content (IWC), it exhibits much more variation with height than LWC (see figure 13.c). It increases with altitude, starting at 2 km and growing up to 7 km for colder months (winter and spring) and up to 8 km for warmer months (fall and summer). Winter and fall follow practically the same comportment, as well summer follows the same behavior of fall up to almost 8 km. Above this level, summer has the largest amount of IWC, with a peak of 15 mg m^{-3} around 11 km. However, ice mass is generally greater at middle and high altitudes, as expected. The right panel of Figure 13.c shows significant values of IWC of about 20-40 mg m^{-3} between 4 and 10 km, except for July. As for the LWC, July also presented elevated IWC above 10 km (more than 60 mg m^{-3}), complementing our hypotheses that these clouds can be classified as cumulus clouds. Cumulus clouds are the first stage of cumulonimbus clouds and can have considerable amounts of ice, especially at their top. In addition, these clouds have a considerable thickness as those observed for mixed phase clouds (some of them can reach 2 km of thickness). Furthermore, the bottom panel of Figure 13.c shows that median values of IWP are comparable to the LWP. However, they are much less variable, which can be seen by the smaller differences between the median and the mean monthly IWP. In general, typical values are below 10 g m^{-2}, being slightly larger in April and summer, when there are more observations.

In terms of effective radius, their size follows a different function of the height. The seasonal profiles (on the left panel of Figure 13) show a smooth decrease of effective radius with height up to 8 km, and a faster decrease above this layer. In the entire column, summer has a slightly larger radius, especially at 11 km, reaching 30 μm, whereas other seasons showed smaller radius at this level. The monthly analysis (see Figure 13) undoubtedly shows that higher radii are at low levels, except in July where it's possible to find r_{ice} larger than 50 μm above 10 km. However, the gradient of colors shows that normally, ice clouds have larger ice particles the closer to the surface, where there is low amounts of ice water content (IWC). In addition, differently from water droplets, which have shown monthly distributions of r_{liq} with the median above the mode, r_{ice} shows a median below the mode (see bottom panel of Figure 13). Overall, the median is fluctuating around 45 μm, with the mode above this value, indicating high probability to find elevated radius.

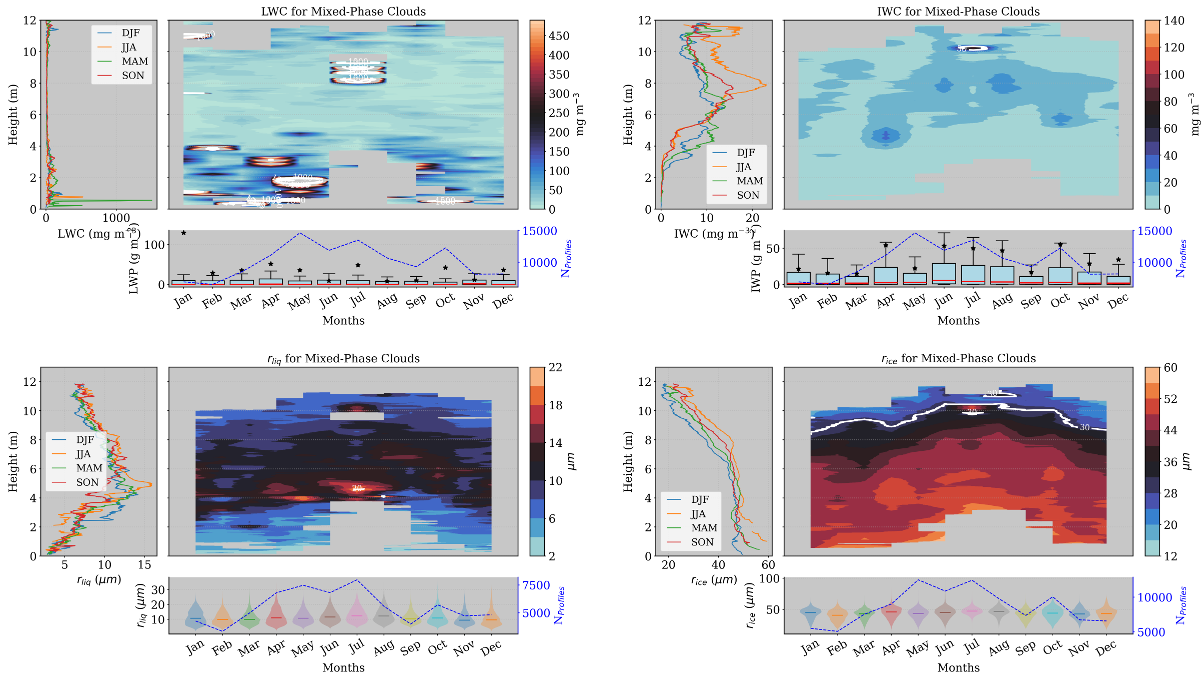


Figure 13. Same as Figure 9, but for mixed-phase clouds.Seasonal and monthly average of ice water content (IWC) and ice water path (IWP) for mixed-phase clouds. Left panel shows the seasonal median profiles of IWC. Right panel shows the monthly median profiles of IWC and monthly box-plot of IWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

Similarly, the seasonal and monthly evaluation of liquid and ice properties of precipitating mixed-phase clouds is shown in Figure 14.a. In the left panel of this figure, spring exhibited the largest values of LWC, with a sharp increase that surpasses 1000 mg m^{-3} near the surface. Additionally, a secondary, much less pronounced peak is observed at 2 km during this season. At low levels, fall also exhibited elevated LWC values, with a median around 300 mg m^{-3} at 500 m. During this season and also summer, considerable values of LWC are found just above 4 km, whereas winter and spring have LWC close to zero just below 4 km. Thus, warmer and dry months (such as summer and fall) cause liquid particles to form at higher levels. As previously mentioned, the temperature in Granada during summer strongly increases, making it difficult for clouds to form, as they have to reach higher altitudes. Additionally, summer reveals a clear water shortage, especially at low levels. Both in summer and fall, considerable amounts of LWC are present at medium levels in the atmosphere (> 2 and < 4.5 km). This can be easily seen in the left panel of Figure 14.a, where fall (in red) and summer (orange) have almost the same vertical profile, varying around 200 mg m^{-3} between 2 and 4.5 km. The monthly profiles in the right panel of Figure 14.a displays that precipitating mixed-phase clouds generally have LWC at lower levels than non-precipitating mixed-phase clouds (practically zero below 5 km). Additionally, they present a stronger seasonal pattern and larger LWC, which is natural given their strong seasonal pattern in frequency of occurrence. It is worth mentioning that the color scale in left panel of Figure 14.a is the same as in the non-precipitating case in Figure 14.a, right panel. Moreover, these precipitating clouds do not present such high variability with height, consistent with their cloud base distribution, showing much less variability than the non-precipitating clouds. The integrated liquid water path (see bottom panel of Figure 14.a) exhibits high variability in colder months, with medians below 50 mg m^{-2} and averages above 100 mg m^{-3}. In summer, just a few occurrences of these clouds were observed, resulting in less variability, with both medians and averages close to zero.

In terms of r_{liq}, seasonal profiles are quite different. Large size droplets are observed close to the surface on account of drizzle particles below the cloud base. Since these clouds usually have the base height above 2km, their r_{liq} is about 8 μm (see left panel of Figure 14.b). Below this level the liquid droplets are larger during summer, and practically the same among the rest of the seasons. Above this level, summer and fall have the largest droplets (> 4 km and <6 km), with maximum sizes at close to 4.5 km. Spring also has the maximum close to this layer, although their radii size is smaller and comparable with the winter maximum at 4 km. Around this layer and also shortly above, the left panel of Figure 14.b indicates larger droplets in January, February and April. For summer, larger values of r_{liq} can be clearly identified around 2 km in August (30 μm) and at 4.5 km in July (20 μm). For spring, droplet sizes are larger close to 5 km, with values slightly smaller than 20 μm. Bear in mind that we are not considering droplets sizes below 2 km, since they are probably related to rain pixels, but classified as drizzle droplets by Cloudnet. In general, the monthly distribution of r_{liq} shows a median around 10 μm, reaching 15 μm in July, August and September.

Following, the left panel of Figure 14.c reveals that the ice amount inside these clouds is found all above 3 km, and exhibit different seasonal profiles. Typically, IWC profiles for summer and spring are similar up to an altitude of 9 km. Just above this level, summer profiles exhibit a distinct peak, exceeding 200 mg m^{-3} of ice. Below this level, spring and winter have similar profiles with less amount of ice. These seasons have a maximum of 22 (winter) and 33 mg m^{-3} (spring) at 4km, respectively. In addition, it is possible to observe a large layer during colder months (winter-spring) with small values fluctuating around 15 mg m^{-3} from 5.5 to 7.5 km. On the other hand, summer and fall presented a maximum of 40 mg m^{-3} at 7 km and 50 mg m^{-3} at 5 km, respectively. Above 8 km, the peak observed in summer is related with the highest values of IWC in June as can be seen in the right panel of Figure 14.c. However, this month has the smaller data availability (see the blue line in the bottom panel), indicating less statistical confidence. Since the entire summer has small data availability, this maximum is reproduced in the vertical profile on the left panel. In fall, the highest values of IWC occur in November, just above 5 km, and ranging from 60 to 100 mg m^{-3}. For winter, the maximum IWC occurs at 5 km in January, reaching more than 120 mg m^{-3}. However, since we cannot observe this peak on the winter profile (on the left panel), January represents just a particular case rather than typical winter conditions. Particularly, July, August, and September have the most comparable largest amounts of IWC in the atmosphere. Consequently, the 25th, 50th, and 75th percentiles during these months are relatively higher, with 25% of the data showing IWC values exceeding 200 mg m^{-2}.

As for the IWC, the r_{ice} have a clear seasonal pattern. It suggests that in general, the ice content has a stronger seasonal pattern than liquid content. The left panel of Figure 14.d shows nearly constant r_{ice} around 50 μm up to 8 km. Above this level, for the warm months, particle sizes decrease in fall while exhibit a sharp maximum at 9.5 km (60 μm) in summer, coinciding with the maximum of IWC. In contrast, winter and spring have similar r_{ice} values to those in summer and fall up to 4 km, after which the sizes decrease with height, reaching 20 μm at 10 km. It is also clear by monthly evaluation that r_{ice} decreases with height, as can be seen on the right panel of Figure 14.d. Particularly during summer and fall, particle sizes exhibit less variability with height, ranging around 50-55 μm. Conversely, winter and spring show variability between 45 and 55 μm. Despite this variability of r_{ice} with height, the monthly distribution shows that within these clouds, r_{ice} is generally around 55 μm.

As for the IWC, the r_{ice} has a clear seasonal pattern. It suggests that, in general, the ice content has a stronger seasonal pattern than liquid content. The left panel of Figure 14.d shows nearly constant r_{ice} around 50 μm up to 8 km. Above this level, during the warm months, particle sizes decrease in fall while exhibiting a sharp maximum at 9.5 km (60 μm) in summer, coinciding with the maximum of IWC. In contrast, winter and spring have similar r_{ice} values to those in summer and fall up to 4 km, after which the sizes decrease with height, reaching 20 μm at 10 km. It is also clear by monthly evaluation that r_{ice} decreases with height, as can be seen on the right panel of Figure 14.d. Particularly during summer and fall, particle sizes exhibit less variability with height, ranging around 50-55 μm. Conversely, winter and spring show variability between 45 and 55 μm. Despite this variability of r_{ice} with height, the monthly distribution shows that within these clouds, r_{ice} is generally around 55 μm.

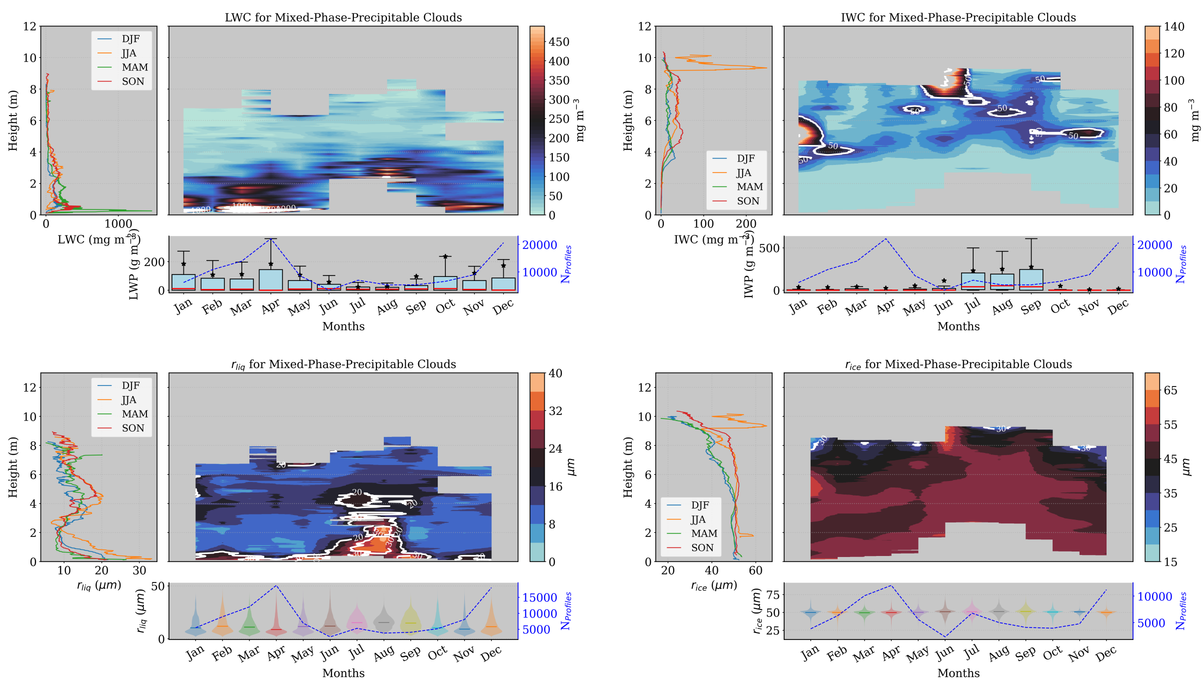


Figure 14. Same as Figure 13, but for precipitating mixed-phase clouds.

Overall, Mixed-phase clouds exhibit less liquid water content (LWC) than liquid clouds. Typically, the LWC for mixed phase clouds is concentrated at low levels, with some exceptions in July, where high LWC values are observed above 8 km. The droplet effective radius in mixed-phase clouds is larger at mid-levels, while the ice effective radius is only large in mid-levels during winter and spring, and larger at 8 km during fall and summer. Unlike this pattern, the ice effective radius in these clouds slightly decreases with height up to 6 km, where the rate of decrease intensifies. Precipitating mixed-phase clouds exhibit a stronger seasonal profile, with higher LWC, particularly in spring and fall at lower levels due to the drizzle presence. They also show significantly more IWC in summer and early fall. The droplet effective radius (r_{liq}) in precipitating mixed-phase clouds is typically higher than in non precipitating mixed-phase clouds, especially near the surface and at mid-levels, due to the presence of drizzle particles at low levels. In terms of the ice effective radius (r_{ice}), these clouds display a clear seasonal variation, with higher r_{ice} values in summer and fall above 4 km. Additionally, precipitating clouds show larger IWC and r_{ice} values at 9 km in summer, suggesting a possible association with cumulus clouds.

5.3.3 Ice Clouds

High amounts of ice during winter create a strong seasonal pattern of IWC for ice clouds. In the left panel of Figure 15, a peak of more than 400 mg m^{-3} of IWC is observed at 1 km. Despite the fact that non-precipitating ice clouds are usually related to cirrus clouds (located at high levels), winter shows low-level ice clouds with much more IWC. However, they represent a minor portion of ice clouds, since we already observed that only 25

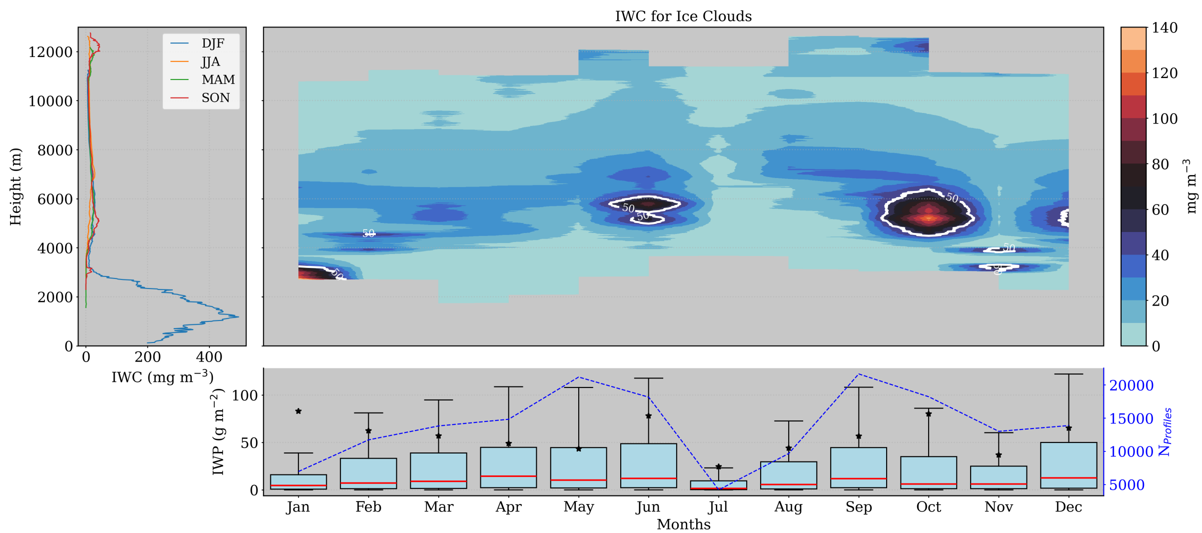


Figure 15. Seasonal and monthly average of ice water content (IWC) and ice water path (IWP) for ice clouds. Left panel shows the seasonal median profiles of IWC. Right panel shows the monthly median profiles of IWC and monthly box-plot of IWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

For the ice effective radius, the largest sizes can be found below 6 km as can be seen in the left panel of Figure 16. Inside this layer, r_{ice} is normally fluctuating around 50 μm, being greater below 3 km only in winter, relating to the snowfalls in January of 2020 and 2021. Above 6 km, ice particle size diminishes, indicating a larger cloud radius at lower levels, a pattern also observed in mixed-phase clouds. Typically, ice particles are larger at low and mid-levels. In the case of cirrus clouds situated above 10 km, the ice effective radius is between 25 and 30 μm, except in winter. These peaks at 11-12 km are coinciding with the IWC.

At this layer (10-12 km), the right panel of Figure 16 display the months responsible for these peaks, which is June, August, October and November (denoted by the white line, contouring regions with more than 30 μm). The monthly distributions in the bottom panel reveals a stable median ice particle size of approximately 38 μm throughout the year, indicating that this value is generally representative when considering all radii within the atmosphere for this cloud type. However, January presents a distribution with a new mode around 50 μm, as expected due to the snowfall events already discussed.

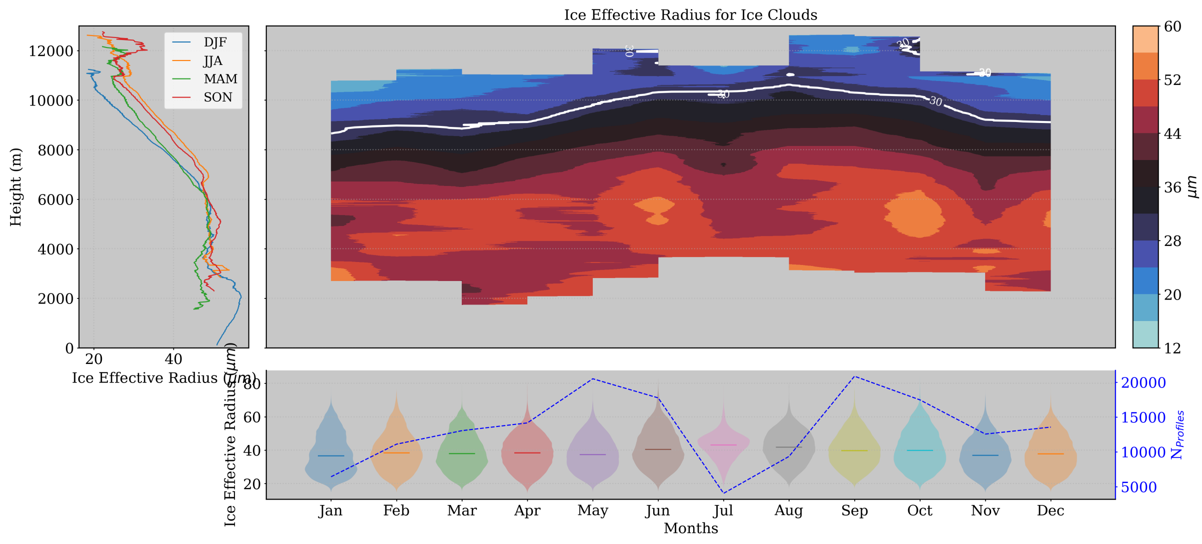


Figure 16. Seasonal and monthly statistics on ice effective radius (r_{ice}) distribution for ice clouds. Left panel shows the seasonal median profiles of ice effective radius. Top-right panel shows the monthly profiles of the median r_{ice}, while the bottom-panel displays the size distribution of r_{ice} by months.

Precipitating ice clouds are the most frequent at our station, with a well defined seasonal cloud occurrence. These clouds are characterized by their greater thickness and lower cloud base height compared to non-precipitating ice clouds. The median profiles of IWC on the left panel of figure 17 shows similar behavior for summer and fall. Both have a first maximum of 200 mg m^{-3} at 6 km and a second maximum at different altitudes. Summer presents a minor second maxima of about 70 mg m^{-3} around 11 km, and fall has a more pronounced maxima surpassing 200 mg m^{-3} just above 12 km. Moreover, spring and winter have less ice amount and also exhibit similar profiles with only one peak of 100 and 150 mg m^{-3} around 4.5 km, respectively. The right panel of Figure 17 turns into evidence the area exceeding 50 mg m^{-3}, located between 2 and 7 km over the year. This layer is higher during summer and fall, where values are also larger, achieving more than 200 mg m^{-3} in June and September. In addition, July also achieves these values between 9 and 12 km, different from the observed low IWC for non-precipitating ice clouds. However, mixed-phase clouds and principally precipitating ice mixed phase clouds also exhibited peaks of IWC at this level in summer. These values can be associated with cumulus and cumulonimbus clouds observed in summer at the Granada station. Although this season presents a very limited occurrence of these clouds in summer, they are normally related to convective clouds. Specifically, precipitating ice clouds can be related to cumulonimbus clouds, and mixed-phase and precipitating mixed-phase clouds can be related to cumulus clouds or final stage cumulonimbus clouds, respectively. The bottom panel of Figure 17 highlights that these clouds contain significantly higher ice water content compared to other clouds, especially during summer and fall, when the 75th percentile increases from 500 to 1000 mg m^{-2}. However, note that despite these values being larger in summer, precipitating ice clouds already showed less cloud occurrence this season, reducing the outliers and turning the IWP values comparable, increasing the 25/75 percentile interval. On the other hand, these clouds have more frequency of occurrence in other months, reducing the variability. It shows that IWP are normally much smaller throughout the year, specially during spring and fall when they are more frequent.

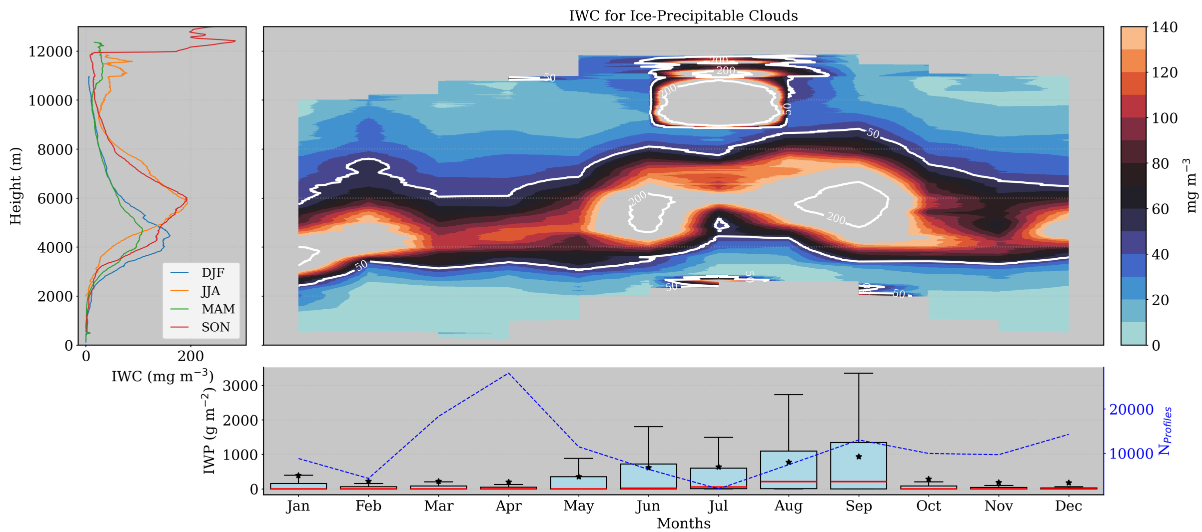


Figure 17. Same as Figure 15, but for ice clouds.

For the ice effective radius (r_{ice}), the left panel of Figure 18 shows particle size slightly growing with height above 1 km for all seasons. Below this level particles are probably related to graupel and are not inside the cloud, then we will consider only the layer above. In general, summer and fall have similar vertical profiles as well as winter has similar profiles as spring up to 10 km, such as the IWC profiles. For winter and spring, r_{ice} increases up to 5 km (reaching 50 μm), whereas in summer and fall, r_{ice} increases up to 7 km (reaching 52 μm). In fact, their maximums are practically the same, but happen at different heights (5 and 7 km). Above their maximums, particle size decreases up to 10 km, when they start to increase again for spring, summer and fall. In spring, r_{ice} starts to increase just above 10 μm and reach at 35 μm at 11 km, while summer maintained r_{ice} at 42 μm between 10 km and 12 km. Then, fall exhibited a sharp increase of r_{ice} at 12 km, reaching a maximum of 65 μm at 13 km. The right panel of Figure 18 is in agreement with the second maximum of r_{ice} observed in summer, showing the largest particles in July. Also, this panel clearly shows the height where sizes start to decrease with seasons, being higher in summer and fall. The second peaks of spring and fall have not enough neighbor data for painting a contour plot, then they are not shown (but they are still there). At last, despite the larger variability with height, the monthly distribution of r_{ice} in the bottom panel of Figure 18 shows a highly variable pattern with a sharp mode at the median position of 50 μm.

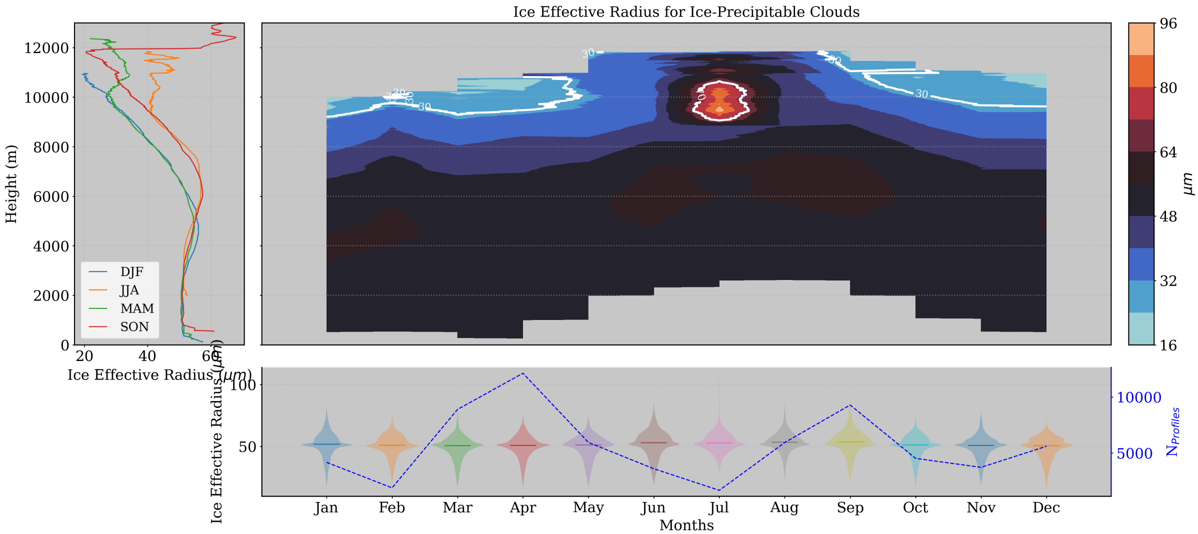


Figure 18. Same as Figure 16, but for mixed-phase clouds.

In general, non precipitating ice clouds exhibit large amounts of ice water content (IWC) at mid-levels (around 5 km), particularly in early summer and fall. In January, most high IWC values were recorded during extreme snowfall events, such as the Filomena and Gloria storms, resulting in high median IWC at low levels during this month. The r_{ice} is also influenced by these snowfalls, showing higher values close to the surface. Typically, r_{ice} ranges around 50 μm up to 6 km, where it starts to decrease, indicating smaller ice particles in higher-level clouds. Precipitating ice clouds display a much stronger seasonal pattern in their physical properties. Summer and fall show a peak of IWC at 6 km, while winter and spring exhibit a less pronounced peak at 4 km. These clouds contain significantly more IWC than non-precipitating ice clouds, and they are markedly different, with greater thickness. Similar to precipitating mixed-phase clouds, we found large amounts of IWC and r_{ice} at high levels in summer (> 8 km). Additionally, they demonstrate clear seasonal differences in r_{ice}, which is greater in summer and fall from 6 km of height.

6 Conclusions

This study presents a statistical analysis of cloud properties using remote sensing ground-based instruments at the AGORA ACTRIS CCRES station, marking the first such analysis in Spain employing the Cloudnet database. Utilizing over half a decade (2018 to 2023) of post-processed data from the Cloudnet consortium, we classified single-layer clouds and retrieved their physical properties using the target classification product. However, this product exhibited some misclassification at our station, necessitating adjustments to avoid misinterpretation of the results. To enhance cloud classification, we compared previous time-based cloud classification methods from recent studies with a novel approach that categorizes clouds into ice (precipitating ice), liquid (precipitating liquid), and mixed-phase (precipitating mixed-phase) clouds. Evaluating this classification method is crucial for understanding the derived cloud properties, as these will be influenced by the chosen method. Consequently, using this new method, we conducted an inter-annual analysis alongside a diurnal cycle analysis of single-layer clouds to assess seasonal variability and identify convective clouds. Additionally, monthly and seasonal analyses of macrophysical and microphysical properties were presented by cloud type to elucidate their differences.

Appendix A: Cloud assessment by time

This method have been used by recent studies such as Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.. Each study has adopted some different criteria to classify single layer clouds, but they all classify cloud layers by profile. Here, we adopted some criteria based on these studies and employed the following method: cloud layers in the atmospheric column were classified through pixel sequence inquiry. These pixel sequences were divided in the following 5 groups according to Răzvan Pîrloagă et al.: liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only an "ice" pixel sequence was encountered. Mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified whether at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Unlike previous studies, whether rain pixels are found with more than 20 pixels (256-1022 m) of distance from cloud base, the clouds are not classified as precipitable, assuming that rain droplets are not falling from these cloud layers. In all cases, for being considered a cloud layer, a minimum sequence of 3 cloud pixels (38-153 m) (Tatiana Nomokonova et al.) must be satisfied, in addition to a minimum sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" to separate multi-layers. Therefore, free hydrometeor layers in the neighborhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on Răzvan Pîrloagă et al., but these values are not currently well defined, their determination has been guided by empirical experience.

Appendix B: Pixel dilatation

The dilation method consists in expanding the ones in the cloud mask to connect the closest groups of hydrometeors, which are likely to belong to the same cloud. The dilatation follows the scheme below, where hydrometeor pixels are represented in blue. Left image represents the original hydrometeor pixels where two groups of clouds were identified. Thus, a dilatation of 1 pixel is applied in all directions, represented by the red pixels in the right image. Therefore, the dilation process will unite these groups with at least 2 pixels of separation distance. Since they are connected, the cloud classification algorithm will save the coordinates of these cloud pixels in just one group (not two, as previously identified).

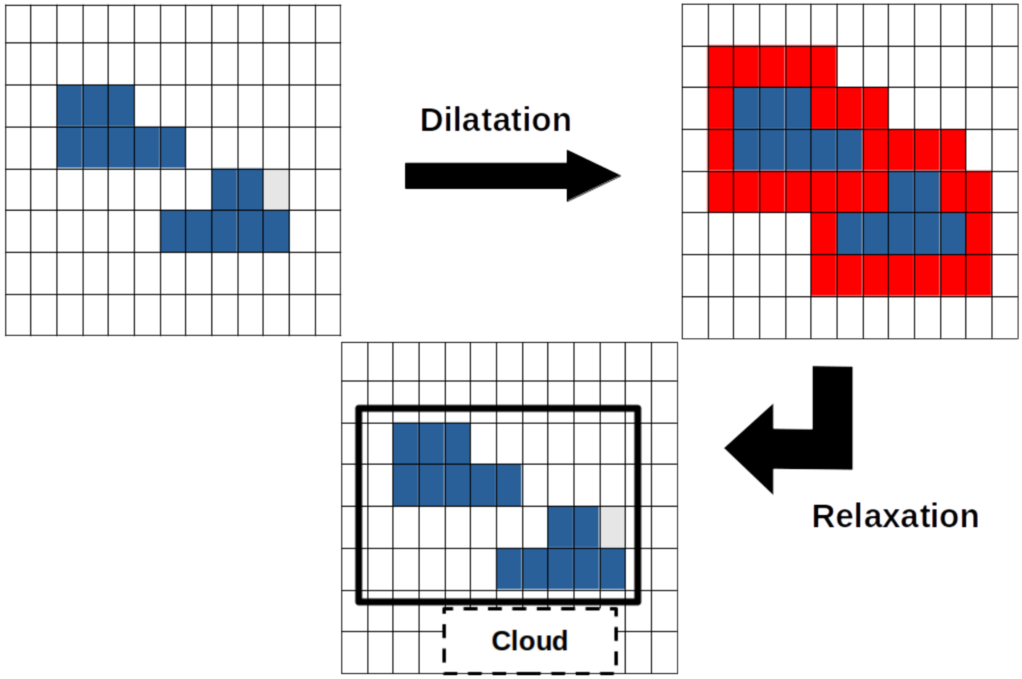


Figure B1. Pixel dilatation and pixel contraction illustration. Left panel shows two hydrometeor groups. Right panel still shows these two clouds and the dilated pixels in red. Since these clouds are connected, the classification algorithm will understand there is only one cloud. Bottom panel shows the result of the pixel contraction, returning cloud pixels to its original position

Next, a contraction process is done to return the coordinates of true hydrometeors to the original position. The contraction will remove the coordinates of fake hydrometeor pixels (pixels in red on the above figure). Afterwards, the classification algorithm can follow its chain and calculate the percentage of each hydrometeor type inside each group to classify different clouds in the cloudnet product.

Acknowledgements. TEXT

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